

SBC 201 Chapter 17 Special Inspections and Tests — Group R-2 High-Rise Dimensional Requirements Matrix (2024)

1. Document metadata and use limitation

- **Project basis:** Riyadh, Saudi Arabia; Group R-2 residential high-rise; an occupied floor is stated to be more than 23 m above the relevant reference level.
- **Deliverable tier:** Project-use matrices in Sections 1–15 (design-check rows, not pasted inventory), plus a coverage summary and unresolved-source register. The full row inventory is not published.
- **Code/source basis:** SBC 201 (2024), Chapter 17, source file Reference\SBC 201 2024\source_reference\Chapter_17 – SPECIAL INSPECTIONS AND TESTS.txt.
- **Extraction audit:** Skill-finetune run. Project-use rows follow the chapter-extract row contract (noun-phrase checks, bold SI values, building-language triggers, named exceptions, check-specific actions). Internal inventory: **156** independently checkable numeric records (**106** Verified, **50** Verify source). Unresolved OCR, flattened tables and page-splits are listed in the register and are not design-release values.
- **Model:** Cursor Grok 4.6.
- **Prepared:** 2026-08-27.
- **Status:** Source-only architectural advisory matrix for design coordination of special inspections, tests and structural observation. It is not a stamped compliance statement, structural-observation report, SCD NOC, SBPS approval, legal opinion, or permission to construct.
- **Outbound-source rule:** No value in this matrix has been imported from SBC 301–306, AISC 341, SDI QA/QC, AWS D1.4, ASTM inspection standards, Section 403, Section 704, Sections 714/715, Section 909, Chapter 16 wind/seismic maps, commentary examples, or the existing chapter summary. Where Chapter 17 sends the user elsewhere, this matrix records the dependency without supplying the outbound value.

Scope and assumptions

1. Group R-2 and high-rise status are project statements, not independently verified classifications.
2. The exact Riyadh AHJ/permit pathway, project stage, fire-strategy status and SCD NOC status are unconfirmed; therefore this matrix does not conclude compliance.
3. Risk Category, Seismic Design Category, basic wind speed *V*, wind Exposure Category, construction type, geotechnical report status, foundation system, SFRM versus mastic/intumescent, smoke-control scope and EIFS use are unconfirmed.
4. Automatic sprinkler protection is not selected. Chapter 17 does not branch NFPA 13 versus 13R. Section 1705.13.6 Item 6 applies where automatic fire sprinkler systems are installed.
5. Building height, storey count, grade plane and mixed-use podium layout are unconfirmed beyond the stated high-rise occupied-floor datum.
6. Seven appended inspection tables are concatenated OCR. No reconstructed continuous/periodic cell or the Table 1705.3 “**7.8 mm**” fillet-weld token is adopted as a design-release value.
7. Commentary numbers are excluded, including fireplace-steel examples, restated **720 N/m²** cladding weight, isolator “two specimen” counts, and Section 403.2.3 SFRM bond (named only in commentary).

2. Legends

Applicability

Status	Meaning
Direct	Expected to govern the stated R-2 tower basis, subject to confirmed geometry and design data.
Conditional	Governs only when the stated feature, material, seismic/wind branch or exception exists.
Not typical	Unrelated occupancy-only rule; omitted from this deliverable unless the gap register already opened that use.
External verification	Chapter 17 points to another section/code/standard, or the project/AHJ basis must be confirmed before use.

Source confidence

Status	Meaning
Verified	Requirement and any stated numeric value were checked against unambiguous mandatory Chapter 17 source text.
Verify source	OCR, flattened table, page-split, or footnote attachment is unresolved. Not a design-release value.

3. Project decision and gap register

Decision / gap	Current project basis	Why it controls Chapter 17 application	Required project action
High-rise confirmation	Occupied floor stated above 23 m; classification not independently verified	1704.6.1 Item 2 structural observation and 1705.18 firestop SI both trigger on high-rise	Issue a signed height/storey datum; keep observation and firestop SI on the SSI until classification is locked
Risk Category	Unconfirmed (R-2 often II, not verified)	RC IV independently requires observation; RC III/IV independently require firestop SI; masonry veneer Level 2 is RC IV only	Structural RDP to assign Risk Category per Section 1604.5 (outbound); do not assume II
Seismic Design Category	Unconfirmed; values live in Chapter 16 / SBC 301	1705.13–1705.14 inspections and tests are SDC-gated; 1704.6.1 Item 3 is SDC E and greater than two stories	Lock SDC on the structural basis of design; show SDC branches on the SSI without importing Chapter 16 maps here
Wind <i>V</i> and exposure	Unconfirmed	Extra wind SI applies in Exposure B at 67 m/sec or greater, or Exposure C/D at 62.5 m/sec or greater	Wind consultant to publish <i>V</i> and exposure; until then retain the two Chapter 17 branches
Construction type	Unconfirmed (steel, concrete, masonry, wood, mass timber)	Selects 1705.2–1705.5 tables and outbound SBC 304/305/306	Freeze primary structure and deck/truss systems on the SSI
Geotech / foundations	Unconfirmed	Soils table, deep-foundation tables, helical piles and 90-percent density exception	Issue approved geotechnical report and foundation type before SSI freeze
SFRM vs intumescent	Unconfirmed	1705.15 sampling/thickness/density/bond versus 1705.16 AWCI 12-B	Fire-protection specification to name the listed system; high-rise 403.2.3 bond is outbound and not imported
Smoke control	Unconfirmed	1705.19 special inspector if a smoke-control system is provided	Fire strategy to lock Section 909 / smokeproof-enclosure scope
EIFS	Unconfirmed	1705.17 SI unless a listed drainage or masonry/concrete-substrate exception applies	Façade specification to confirm EIFS, WRB coating and substrate
Sprinkler presence	Unconfirmed; 13 vs 13R not a Chapter 17 branch	1705.13.6 Item 6 sprinkler-clearance SI applies where automatic sprinklers are installed	Fire engineer to confirm sprinklers; use the 75 mm / flexible-hose branches, not a 13/13R choice

Decision / gap	Current project basis	Why it controls Chapter 17 application	Required project action
Podium mixed use / storage racks	Unconfirmed	Storage-rack SI is SDC D–F and 2400 mm or greater; not a typical apartment fit-out	Open this branch only if warehouse/retail racks appear
NOC / SSI / observation	Unconfirmed	Statement of special inspections is a permit condition; observation does not waive Section 110 or 1705	Engage the structural RDP and approved agencies; align SSI, SBPS and SCD comments before construction

4. Employment, statement of special inspections and submittals

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1703.1	Approved-agency accreditation	Approved agency shall meet Part III of SBC 302 and shall be accredited	Agency performing tests or special inspections	Independence, equipment and personnel in 1703.1.1–1703.1.3; SBC 302 values are not imported	Direct	Name accredited agencies on the SSI; do not let the contractor employ the agency unless the contractor is also the owner	Verified
1703.1.1	Agency independence	Agency shall be objective, competent and independent from the contractor responsible for the inspected work, and shall disclose conflicts to the building official and the RDP in responsible charge	Approved agency	1704.2 Exception 4 allows the contractor to employ the agency only where the contractor is also the owner	Direct	Keep the owner (not the GC) as the employing party on this tower	Verified
1704.2	Owner-employed special inspections	Owner or owner’s authorized agent, other than the contractor, shall employ approved agencies for Section 1705 work, in addition to Section 110 inspections	Permit application under Section 105	1. Minor work or jurisdiction-approved conditions. 2. Group U accessory to residential unless the building official requires SI. 3. Portions under 2211.1.2 CFS light-frame or 2308 conventional light-frame. 4. Contractor-as-owner employment	Direct	Place SI on the high-rise permit path; do not treat the tower as minor work or prescriptive light-frame	Verified
1704.2.3 / 1704.3	Statement of special inspections	Applicant shall submit an SSI as a permit condition; the RDP in responsible charge prepares it	Special inspections or tests required by 1705	SSI not required for 2211.1.2 / 2308 portions; a qualified person approved by the building official may prepare the SSI only where the work was not designed by an RDP	Direct	Issue the SSI with the permit set; do not defer it to construction administration	Verified
1704.3.1	SSI content	Identify materials/systems/work requiring SI or tests; type and extent of each inspection and each test; extra seismic/wind items under 1705.12–1705.14; and whether each inspection is continuous, periodic, or per the referenced-standard notation	Every required SI/test	Wind and seismic identification in 1704.3.2–1704.3.3 when those sections apply	Direct	Schedule each tower SI scope as continuous, periodic or standard-notation on the SSI	Verified
1704.2.4	Inspection reports	Approved agencies keep records, report to the building official and the RDP, flag uncorrected discrepancies before phase completion, and submit a final report at a time agreed before start of work	SI and tests performed	None stated	Direct	Put report routing and the agreed final-report date in the CA/SSI kickoff	Verified
1704.2.5 / 1704.2.5.1	Fabricator special inspection	Shop fabrication of structural, load-bearing or lateral-load-resisting members requires SI during fabrication unless the fabricator is approved; approved fabricator submits a certificate of compliance	Off-site structural fabrication	Approval based on written procedures, QC manuals and periodic auditing	Conditional	Require SI or an approved-fabricator certificate for precast, steel and truss shops	Verified
1704.4	Contractor statement of responsibility	Each contractor for a listed main wind or seismic force-resisting system, designated seismic system, or wind/seismic component shall submit a written statement of responsibility before that work starts	Wind or seismic items listed on the SSI	Acknowledgement of the SSI special requirements only	Conditional	Collect contractor statements before steel/concrete/façade erection on those systems	Verified
1704.5 Items 1–7	Owner certificates and reports	Owner submits fabricator, seismic-qualification, designated-seismic, shotcrete, joist, weldability and SDC B–F mill-test certificates/reports listed in this section, in addition to 1704.2.4 agency reports	Applicable listed materials/systems	Item 7 mill tests apply to ASTM A615 bars used to resist earthquake flexure/axial force in special moment frames, special structural walls or coupling beams in SDC B, C, D, E or F ; outbound SBC 304 clauses are not imported	Conditional	Schedule the listed certificates against actual materials; lock SDC before relying on Item 7	Verified

5. Structural observation

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1704.6 / 1704.6.1 Item 2	High-rise structural observation	Owner shall employ a registered design professional to visually observe representative structural systems, details and load paths; observation does not waive Section 110 or 1705	Structure is a high-rise building	Also required for Risk Category IV , or SDC E and greater than two stories above grade plane, or where the structural RDP or building official requires it	Direct	Appoint the structural observer, file frequency/extent before site visits, and file the close-out statement of unresolved deficiencies	Verified
1704.6.1 Item 3	SDC E multi-storey observation	Structural observation required where the structure is SDC E and greater than two stories above grade plane	SDC E and storey count	Independent of the high-rise trigger; this tower already meets Item 2 if high-rise is confirmed	Conditional	Retain Item 2 as the governing high-rise trigger; Item 3 is additional if SDC E is assigned	Verified

6. Steel, concrete, masonry and wood special inspections

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1705.2.1	Structural steel SI	Special inspections and NDT of structural steel elements shall follow the quality-assurance inspection requirements of SBC 306	Structural steel in the building	Fabrication SI not required where the entire fabrication process has no welding, thermal cutting or heating, with stated material-control procedures; railing systems limited to weld inspection at cantilevered rail-post bases	Direct	List structural steel SI on the SSI as SBC 306 QA; do not import SBC 306 frequencies here	Verified
1705.2.2	Steel deck SI	Cold-formed steel floor and roof deck SI and welding-inspector qualification shall follow SDI QA/QC	Steel floor or roof deck	None stated in this section	Conditional	If metal deck is used, name SDI QA/QC on the SSI without importing that standard's values	Verified
1705.2.3 / Table 1705.2.3	Open-web joist SI	Open-web steel joists and joist girders shall be specially inspected in accordance with Table 1705.2.3	Joists/girders used	Table is concatenated OCR; continuous/periodic <i>X</i> marks are not adopted as design-release values; Note a also points to 1705.13	Conditional	If joists are specified, verify the published table before assigning periodic vs continuous; keep joist certificates per 1704.5 Item 5	Verify source
1705.2.4	Long-span CFS truss bracing	Where cold-formed steel truss clear span is 18 m or greater , the special inspector shall verify temporary installation restraint/bracing and permanent individual truss-member restraint/bracing against the approved truss submittal	CFS truss span ≥ 18 m	None stated	Conditional	Flag any CFS truss ≥ 18 m on framing plans and put bracing SI on the SSI	Verified
1705.3 / Table 1705.3	Concrete construction SI	Concrete SI and tests shall follow this section and Table 1705.3	Concrete construction	Story-limited footing exceptions and on-grade slabs/patios in 1705.3; this high-rise fails the three stories or less gate. Table is concatenated OCR. Item 2b token “maximum 7.8 mm ” fillet weld is not adopted	Direct	Put concrete SI on the SSI; verify published Table 1705.3 frequencies before issuing hold-points; do not repair 7.8 mm from memory	Verify source
1705.3 Ex. 1–2	Low-rise footing SI waiver	Isolated spread footings, and continuous wall footings, of buildings three stories or less above grade plane fully supported on earth or rock may be waived as stated	Buildings three stories or less	Continuous-footing sub-items are flattened (items 3–5 appear un-nested; 17. MPa OCR). Waiver does not describe this high-rise	Conditional	Do not apply the three-storey footing waiver to the tower; verify published exception nesting only if a separate low-rise U/podium structure is claimed	Verify source
1705.3.1	Rebar welding SI	Welding SI and inspector qualification for reinforcing bars shall be in accordance with AWS D1.4	Rebar is welded	Source repeats AWS D1.4 twice; AWS values are not imported	Conditional	If rebar welding is specified, name AWS D1.4 on the SSI and collect 1704.5 Item 6 weldability reports for non-A706 bars	Verified
1705.4	Masonry SI	Masonry SI and tests shall follow the quality-assurance program of SBC 305	Masonry construction	Empirically designed masonry, glass-unit masonry or masonry veneer in RC I, II or III ; masonry fireplaces/heaters/chimneys per 2111–2113	Conditional	If structural or veneer masonry is used, list SBC 305 QA; do not import SBC 305 levels except as 1705.4.1 states	Verified
1705.4.1	RC IV veneer Level 2	Glass-unit masonry or masonry veneer in Risk Category IV shall be inspected to SBC 305 Level 2	RC IV structure with that masonry	Ordinary R-2 is not typically RC IV	Conditional	Apply Level 2 only if RC IV is assigned; otherwise keep 1705.4	Verified
1705.5.1	High-load wood diaphragm SI	High-load diaphragms designed to 2306.2 require SI of panel grade/thickness, adjoining framing size, fastener diameter/length, fastener-line count and spacing/edge margins	2306.2 high-load diaphragm	Prefabricated wood assemblies follow 1704.2.5	Conditional	Typical concrete/steel R-2 towers omit this unless a wood diaphragm is specified	Verified
1705.5.2	Long-span wood truss bracing	Where metal-plate-connected wood truss clear span is 18 m or greater , verify temporary and permanent restraint/bracing against the approved truss submittal	Wood truss span ≥ 18 m	None stated	Conditional	Same 18 m gate as CFS; apply only if long-span wood trusses exist	Verified
1705.5.3 / Table 1705.5.3	Mass-timber SI	Mass timber in Types IV-A, IV-B and IV-C shall follow Table 1705.5.3 (source cite printed as Table 1/705.5.3)	Mass-timber construction types	Flattened table cells are not adopted	Conditional	Conventional R-2 high-rise is not mass timber unless that type is selected; verify published table if IV-A/B/C is used	Verify source

7. Soils and deep foundations

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1705.6 / Table 1705.6	Soils SI and tests	Existing site soils, fill placement and load-bearing requirements shall be inspected and tested per this section and Table 1705.6, using the approved geotechnical report and construction documents	Shallow foundations / compacted fill	Table is concatenated OCR. Where SBC 303 does not require fill-placement reporting, in-place dry density shall be not less than 90 percent of maximum dry density at optimum moisture (ASTM D1557)	Direct	Put soils SI on the SSI; use the 90-percent density check only on the SBC 303 non-reporting branch; do not import commentary 300 mm fill limits	Verify source
1705.7 / Table 1705.7	Driven deep-foundation SI	Driven elements shall be inspected and tested per Table 1705.7 against the geotechnical report and documents	Driven piles	Flattened continuous <i>X</i> marks are not design-release; steel/concrete/specialty rows point back to 1705.2 / 1705.3 / the SSI	Conditional	If driven piles are selected, verify the published table and keep driving records per element	Verify source
1705.8 / Table 1705.8	Cast-in-place deep-foundation SI	Cast-in-place elements shall be inspected and tested per Table 1705.8	Bored piles / caissons	Flattened table; concrete row points to 1705.3	Conditional	If CIP piles are selected, verify the published table and record diameters, lengths and grout/concrete volumes	Verify source
1705.9	Helical-pile SI	Continuous special inspections during helical-pile installation, recording equipment, pile dimensions, tip elevations, final depth, final installation torque and RDP-required data	Helical piles	Geotechnical report and documents govern compliance	Conditional	Use only if helical piles are specified; keep continuous field logs	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1705.10	Deep-foundation integrity tests	Where structural integrity is reasonably in doubt, an engineering assessment with defect tests per ASTM D4945, D5882, D6760 or D7949, or other approved method, is required	Doubt as to element integrity	Test method values are not imported	Conditional	Trigger integrity testing from driving/drilling records, not as a default extra on every pile	Verified

8. Wind-resistance special inspections

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1705.12	Wind SI geographic gate	Extra wind-resistance SI in 1705.12.1–1705.12.3 is required in Exposure B where <i>V</i> is 67 m/sec or greater , and in Exposure C or D where <i>V</i> is 62.5 m/sec or greater , unless 1704.2 exceptions apply	Wind speed and exposure from the structural basis of design	Do not assume Riyadh <i>V</i> or exposure in this chapter	Conditional	Publish <i>V</i> and exposure; if either gate is met, identify MWFRS and cladding on the SSI per 1704.3.3	Verified
1705.12.1	Wood MWFRS wind SI	Continuous SI during field gluing of MWFRS elements; periodic SI of nailing, bolting, anchoring and other fastening of wood MWFRS elements	1705.12 wind gate and wood MWFRS	Not required where lateral resistance is structural sheathing and specified panel-edge fastener spacing is more than 100 mm on center	Conditional	Typical concrete/steel R-2 omits wood MWFRS; if used, apply the 100 mm spacing exception only where documented	Verified
1705.12.2	CFS light-frame wind SI	Periodic SI of MWFRS welding and of screw, bolt, anchor and other fastening	1705.12 wind gate and CFS MWFRS	Not required for gypsum/fiberboard sheathing, or wood-panel/steel-sheet sheathing on only one side with panel-edge spacing more than 100 mm o.c.	Conditional	Apply only to CFS lateral systems under the wind gate	Verified
1705.12.3	Cladding and roof wind SI	Periodic SI of fastening of roof covering/deck/framing connections, and of exterior wall covering and wall connections to roof and floor diaphragms and framing	1705.12 wind gate	None stated beyond 1704.2	Conditional	If the <i>V</i> /exposure gate is met, add façade and roof-fastening SI to the envelope specification	Verified

9. Seismic-resistance special inspections and testing

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1705.13 Exception	Low-seismic SI waiver	1705.13.1–1705.13.9 SI is not required for light-frame with <i>S_DS</i> not more than 0.5 and height not more than 10.5 m ; or reinforced masonry/concrete with <i>S_DS</i> not more than 0.5 and height not more than 7.5 m ; or detached one- or two-family dwellings not exceeding two stories without listed SBC 301 12.3 irregularities	Structures meeting a listed exception	A high-rise exceeds both 10.5 m and 7.5 m height limits; the dwelling exception is not this occupancy	Conditional	Do not apply the 1705.13 height waivers to this tower; still lock SDC/ <i>S_DS</i> on the structural basis of design	Verified
1705.13.1.1	Steel SFRS seismic SI	Structural steel seismic force-resisting systems in SDC B, C, D, E or F shall be inspected to AISC 341 quality assurance	Steel SFRS and SDC B–F	SDC B or C where the SBC 301 Table 12.2-1 “steel systems not specifically detailed for seismic resistance, excluding cantilever column systems” <i>R</i> row is used; SDC D, E or F where SBC 306 detailing is permitted by SBC 301 Table 15.4-1. Do not take <i>R</i> = 3 from commentary	Conditional	After SDC is locked, list AISC 341 SI unless a stated exception is demonstrated; do not import AISC 341 frequencies	Verified
1705.13.1.2	Steel collector/chord SI	Structural steel elements in the SFRS other than 1705.13.1.1 (struts, collectors, chords, foundation elements) in SDC B–F follow AISC 341 QA	Those steel elements	SDC B or C with <i>R</i> of 3 or less ; SDC D, E or F where detailing other than AISC 341 is permitted by SBC 301 Table 15.4-1	Conditional	Identify collectors/chords on the SSI separately from the main steel SFRS	Verified
Unnumbered — page-split after 1705.13.1.2	Wood SFRS seismic SI	For SFRS of structures assigned to SDC C, D, E or F : continuous SI during field gluing; periodic SI of nailing, bolting, anchoring and other fastening of wood SFRS elements	Wood SFRS in SDC C–F	Not required where lateral resistance is structural sheathing and panel-edge spacing is more than 100 mm on center. Published subsection number is missing in the extract	Conditional	Verify the printed 1705.13.2 number; apply only if a wood SFRS exists	Verify source
1705.13.3	CFS SFRS seismic SI	Periodic SI of SFRS welding and of screw/bolt/anchor/other fastening in SDC C, D, E or F	CFS light-frame SFRS	Gypsum/fiberboard sheathing; or one-sided wood-panel/steel-sheet sheathing with edge spacing more than 100 mm o.c.	Conditional	Apply only to CFS lateral systems in SDC C–F	Verified
1705.13.4	Designated seismic systems	In SDC C, D, E or F , the special inspector shall examine designated seismic systems requiring qualification under SBC 301 13.2.2 and verify label, anchorage and mounting against the certificate of compliance	Designated seismic systems	SBC 301 qualification criteria are not imported	Conditional	List designated seismic equipment on the SSI and collect 1704.5 Item 3 certificates	Verified
1705.13.8	Seismic isolation SI	Periodic SI during fabrication and installation of isolator units and energy-dissipation devices in seismically isolated structures in SDC B–F	Seismic isolation used	None stated	Conditional	Typical R-2 does not use isolation; open only if specified	Verified
1705.13.9	CFS special bolted moment frames	Periodic SI of CFS special bolted moment-frame installation in the SFRS of SDC D, E or F	That SFRS in SDC D–F	None stated	Conditional	Apply only if that system is selected	Verified
1705.14.1.1–1705.14.1.2	Steel seismic NDT	NDT of steel SFRS and of other steel SFRS elements in SDC B–F follows AISC 341 QA	Steel SFRS / steel SFRS elements	Same SDC/ <i>R</i> /SBC 306 exception pattern as 1705.13.1.1–1705.13.1.2, including <i>R</i> of 3 or less on elements in SDC B or C	Conditional	Pair NDT with the matching SI exception branch; do not import AISC 341 NDT rates	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1705.14.2	Nonstructural seismic qualification	In SDC B–F , where SBC 301 13.2.1 Item 2 seismic qualification is used, the RDP shall specify qualification requirements on the construction documents and certificates of compliance shall be submitted per 1704.5	Qualification path chosen instead of project-specific design	SBC 301 / AC 156 values are not imported	Conditional	If qualification (not design) is chosen, put the parameters on drawings and collect certificates	Verified
1705.14.3	Designated-system seismic tests	1704.5 Item 3 requires certificates of compliance for designated seismic systems in accordance with Section 1705.14.3; that subsection is not present in the supplied extract (1705.14.2 is followed by 1705.14.4)	Designated seismic systems	No value is adopted from commentary or outbound SBC 301	External verification	Verify published 1705.14.3 before specifying designated-system tests	Verify source
1705.14.4	Isolation-system tests	Seismic isolation systems in SDC B–F shall be tested in accordance with Section 17.8 of SBC 301	Seismically isolated structures	Commentary “two specimens” is not adopted	Conditional	Name SBC 301 17.8 on the SSI if isolation is used; do not import specimen counts	Verified

10. Architectural and MEP seismic components

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1705.13.5	Cladding and veneer SI	Periodic SI of erection and fastening of exterior cladding, interior and exterior nonbearing walls, and interior and exterior veneer in SDC D, E or F	Those components in SDC D–F	1. Height 9 m or less above grade or walking surface. 2. Exterior cladding and interior/exterior veneer 250 N/m² or less . 3. Interior nonbearing walls 750 N/m² or less . Commentary 720 N/m² is not used	Conditional	After SDC is locked, SI tower façades above 9 m unless a weight exception is documented; use 750 N/m² for interior walls	Verified
1705.13.5.1	Access-floor anchorage SI	Periodic SI of access-floor anchorage in SDC D, E or F	Raised access floors	None stated	Conditional	Apply to podium IT/MEP access floors if SDC D–F	Verified
1705.13.6 Items 1–4	Emergency power and hazardous MEP SI	Periodic SI of: emergency/standby electrical-equipment anchorage in SDC C–F ; other electrical-equipment anchorage in SDC E or F ; hazardous-material piping and associated mechanical units in SDC C–F ; hazardous-material ductwork in SDC C–F	Those MEP systems and SDC	None stated beyond 1704.2	Conditional	Put life-safety electrical gear and any hazardous MEP on the SSI once SDC is known	Verified
1705.13.6 Item 5	Vibration-isolation clearance SI	Periodic SI of vibration-isolation systems in SDC C–F where documents require a nominal clearance of 7 mm or less between equipment support frame and restraint	Tight isolation clearance specified	Not required where the 7 mm clearance is not specified	Conditional	If isolated plant uses the reduced clearance, add SI; do not import SBC 301 force reductions	Verified
1705.13.6 Item 6	Sprinkler seismic clearance	Where automatic sprinklers are installed in SDC C–F , periodic SI shall verify either SBC 301 13.2.3 clearances (not imported) or a nominal clearance of not less than 75 mm between sprinkler drops/sprigs and non-supporting structure, attached equipment and other piping	Automatic sprinklers present	Flexible sprinkler hose fittings: clearance SI not required. Chapter 17 does not choose NFPA 13 vs 13R	Conditional	Coordinate sprinkler drops with structure and MEP; use the 75 mm branch or flexible hose; do not assume 13 vs 13R here	Verified
1705.13.7 / Table 1705.13.7	Storage-rack SI	Steel storage racks and steel cantilevered racks 2400 mm or greater in height in SDC D, E or F shall have periodic SI per Table 1705.13.7	Tall racks in SDC D–F	Flattened table cells are not adopted; not a typical apartment fit-out	Conditional	Open only if podium warehouse/retail racks meet the height and SDC gates; then verify the published table	Verify source

11. Sprayed fire-resistant materials and intumescent coatings

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1705.15 (heading number page-split)	SFRM SI timing and sample cap	SFRM on floor/roof/wall assemblies and structural members shall be inspected and tested to 1705.15.1–1705.15.6 during construction, with an additional visual inspection after rough electrical, sprinkler, mechanical, plumbing and ceiling-suspension installation and before concealment; required sample size shall not exceed 110 percent of the referenced-standard sample in 1705.15.4.1–1705.15.4.9	SFRM used for the fire-resistance design	Charging heading number is missing; 1705.15.1–1705.15.6 follow. Section 403.2.3 high-rise bond named only in commentary is not imported	Direct	If steel fireproofing is SFRM, schedule in-progress and post-rough visual SI and cap extra samples at 110 percent	Verified
1705.15.1	SFRM verification items	SI and tests shall include substrate condition, thickness, density in kg/m³, bond strength adhesion/cohesion, and finished-application condition	SFRM application	Listing and fire-resistance rating govern; kg/m³ is the published density unit, not a numeric limit	Direct	Put the five checks in the fireproofing specification	Verified
1705.15.4	SFRM thickness tolerance	Not more than 10 percent of thickness measurements may be less than the approved fire-resistance design thickness, and none shall be less than 1705.15.4.1	SFRM thickness survey	1705.15.4.1 floors apply to individual readings	Direct	Reject lots that miss design thickness on more than 10 percent of readings or that violate the individual floor	Verified
1705.15.4.1	Individual thickness floor	For design thickness 25 mm or greater , minimum individual thickness is design thickness minus 6.4 mm . For design thickness less than 25 mm , minimum individual thickness is design thickness minus 25 percent . Determine thickness per ASTM E605	Individual SFRM readings	Sampling locations in 1705.15.4.2–1705.15.4.3	Direct	State both floors in the fireproofing QA sheet; do not import ASTM E605 sample rates except as this chapter modifies them	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1705.15.4.2	Assembly thickness sampling	Not less than four measurements for each 93 m² of sprayed area, or portion thereof, in each story, per ASTM E605	Floor, roof and wall SFRM	None stated	Direct	Sample every 93 m² per story, not the larger ASTM default	Verified
1705.15.4.3–1705.15.4.4	Deck thickness patch	Thickness from a 300 mm by 300 mm square; not fewer than four measurements, located symmetrically; fluted decks include valley, crest and sides and report the average	Cellular or fluted deck SFRM	None stated	Conditional	Use the 300 mm patch on metal-deck fireproofing	Verified
1705.15.4.5–1705.15.4.9	Member thickness sampling	Thickness testing on not less than 25 percent of structural members on each floor; measurements at each end of a 300 mm length: beams/girders nine locations, joists/trusses seven , wide-flanged columns 12 , HSS/pipe columns not fewer than four	SFRM on structural members	None stated	Direct	Put member-type location counts in the QA procedure	Verified
1705.15.5	SFRM density sampling	Density not less than the approved fire-resistance design; ASTM E605; not less than one sample per 230 m² (or portion) of sprayed assembly area in each story, and not less than one sample of each structural-member type per 230 m² of floor area (or portion) in each story	SFRM density	None stated	Direct	Sample density on the 230 m² grid, separate from thickness patches	Verified
1705.15.6 / 1705.15.6.1–1705.15.6.2	SFRM bond strength	Cohesive/adhesive bond not less than 7.2 kN/m² by ASTM E736 field test; not less than one sample per 230 m² of sprayed assembly area per story, and not less than one sample of each member type per 230 m² of floor area per story	Cured SFRM	Primer/paint/encapsulant bond tests in 1705.15.6.3 where listing has not established bond. Do not import Section 403.2.3 high-rise bond from commentary	Direct	Specify 7.2 kN/m² as the Chapter 17 floor; ask the fire engineer whether 403 imposes a stricter outbound value	Verified
1705.16	Mastic and intumescent SI	SI and tests of mastic and intumescent fire-resistant coatings on structural elements and decks shall follow AWCI 12-B , based on the listed fire-resistance design, during construction, with additional visual inspection after rough MEP/sprinkler and before concealment	Intumescent/mastic fireproofing	AWCI 12-B values are not imported	Conditional	If intumescent is used instead of SFRM, name AWCI 12-B on the SSI	Verified

12. Fire-resistant penetrations and joints

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1705.18	Firestop and joint SI trigger	In high-rise buildings, in buildings assigned to Risk Category III or IV , or in fire areas containing Group R occupancies with occupant load greater than 250 , SI is required for through-penetrations, membrane-penetration firestops, fire-resistant joint systems and perimeter fire-containment systems tested and listed to 714.4.1.2, 714.5.1.2, 715.3.1 and 715.4	High-rise, or RC III/IV, or Group R fire area OL > 250	The three triggers are independent. Commentary that concatenates "high-rise assigned to RC III or IV" is not used	Direct	Put firestop/joint/perimeter-containment SI on the SSI because the building is stated as high-rise; the OL > 250 branch is additional, not a substitute	Verified
1705.18.1	Penetration firestop inspection standard	Listed penetration firestops under 714.4.1.2 and 714.5.2 shall be inspected by an approved agency in accordance with ASTM E2174	1705.18 trigger met	ASTM E2174 procedures/values are not imported	Direct	Name ASTM E2174 on the SSI and keep listed-system IDs on the firestop schedule	Verified
1705.18.2	Joint and perimeter inspection standard	Listed fire-resistant joint systems under 715.3.1 and 715.4 shall be inspected by an approved agency in accordance with ASTM E2393	1705.18 trigger met	Includes curtain-wall/floor void systems discussed in the clause; ASTM E2393 values are not imported	Direct	Name ASTM E2393 for joints and perimeter containment; coordinate curtain-wall head-of-wall details	Verified

13. Smoke-control testing

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1705.19 / 1705.19.1	Smoke-control special inspector	Smoke-control systems shall be tested by a special inspector: (1) during duct erection and before concealment for leakage testing and device-location records; (2) prior to occupancy after sufficient completion for pressure-difference testing, flow measurements and detection/control verification	Smoke-control system provided	Section 909 quantitative leakage/pressure values are not imported	Conditional	If the fire strategy includes smoke control or smokeproof enclosures, appoint the 1705.19 inspector at the two stated stages	Verified
1705.19.2	Smoke-control inspector competence	Approved agencies for smoke-control testing shall have expertise in fire-protection engineering, mechanical engineering and certification as air balancers	1705.19 testing	None stated	Conditional	Check agency CVs for all three competences before appointment	Verified

14. EIFS, load tests and exterior window/door assemblies

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1705.17	EIFS special inspection	Special inspections are required for all EIFS applications	EIFS used	1. EIFS over a water-resistive barrier with a means of draining moisture to the exterior. 2. EIFS over masonry or concrete walls	Conditional	If barrier EIFS on stud walls is used, keep SI; drained EIFS or masonry/concrete substrate may take the exception	Verified
1705.17.1	WRB coating SI	A water-resistive barrier coating complying with ASTM E2570 requires SI of that coating when installed over a sheathing substrate	E2570 coating on sheathing	None stated	Conditional	If the Chapter 14 drainage EIFS option uses an E2570 coating, add coating SI	Verified
1705.20	Mass-timber sealant SI	Periodic SI of sealants or adhesives required by Section 703.7 applied to mass-timber elements designated on the approved construction documents	Mass timber with 703.7 sealant	703.7 values are not imported	Conditional	Apply only on IV-A/B/C mass-timber projects	Verified
1708.2.2	Unspecified in-situ load test	Where no referenced load-test procedure exists: test load at least the specified factored design loads (non-SFRS); static load left 24 hours ; recovery not less than 75 percent of maximum deflection within 24 hours after removal; no evidence of failure; deflection not beyond 1604.3 (outbound)	In-situ test of existing/doubtful construction	Duration-sensitive materials adjusted; dynamic components held for a period consistent with function	Conditional	Use only if capacity is in doubt and no material-standard test applies; do not import 1604.3 limits	Verified
1709.3.1	Unspecified preconstruction load test	Superimposed test load not less than two times the superimposed design load, held 24 hours ; recovery not less than 75 percent within 24 hours after removal; then reload to failure or two and one-half times the deflection-limit load or two and one-half times the superimposed design load; allowable load is the least of the deflection-limit load, failure load divided by 2.5 , and maximum applied load divided by 2.5	Construction not designable by approved analysis / no standard test	1709.3.2 sends deflection limits to 1604.3 (not imported)	Conditional	Use only for non-standard assemblies; keep the 2× / 2.5× / 75 percent / 24-hour sequence	Verified
1709.5	Window/door ASD conversion	For exterior windows and doors tested under 1709.5.1 or 1709.5.2, required design wind pressures from SBC 301 may be converted to allowable stress design by multiplying by 0.6	Window/door wind-pressure rating	Alternate-size analysis/testing exception under AAMA 2502; SBC 301 pressures are not imported	Direct	Convert strength-level façade pressures by 0.6 only for these assembly tests; do not invent V	Verified
Unnumbered — following 1709.5 (commentary indicates 1709.5.1)	Labeled window/door assemblies	Exterior windows and sliding doors tested and labeled to AAMA/WDMA/CSA 101/I.S.2/A440; side-hinged doors labeled to that standard or 1709.5.2; labeled products not subject to 2403.2 and 2403.3	Exterior windows and doors	Published 1709.5.1 heading is missing in the extract	Direct	Specify labeled AAMA/WDMA/CSA assemblies; verify the printed subsection number	Verify source
1709.5.2	ASTM E330 assembly test	Assemblies not under 1709.5.1 shall be tested to ASTM E330; glass complies with 2403; each assembly tested for 10 seconds at 1.5 times the Chapter 16 design pressure	Non-AAMA labeled path	Chapter 24 glass thickness not imported	Conditional	If the ASTM path is used, put 10 s / 1.5× in the test specification	Verified
1709.5.2.1	Garage and rolling doors	Tested to ASTM E330 or ANSI/DASMA 108 and meeting DASMA 108 pass/fail; permanent label with manufacturer, model, positive/negative design wind pressure, installation-drawing reference and test standard	Garage/rolling doors	Typical R-2 podium parking doors	Conditional	Require labeled DASMA/E330 doors on parking entries	Verified
1709.5.3.1	Windborne-debris shutters	Impact-protective systems tested to ASTM E1886 and E1996 for impact and ASTM E330 for design wind pressure; SBC 301 pressures multiplied by 0.6 for this section; permanent label per 1703.5.4	Buildings in windborne-debris regions (1609.2, outbound)	Riyadh debris-region status is not determined in this chapter	Conditional	Open only if 1609.2 places the site in a debris region; then use the 0.6 conversion and labeled shutters	Verified

15. Project-use controls

1. Use **Verified** rows for initial SSI and specification scoping after the row trigger and branch are confirmed.
2. Treat every **Verify source** row (flattened tables, 7.8 mm weld token, **17. MPa** OCR, missing 1705.13.2 / 1705.14.3 / 1709.5.1 headings) as a design hold point; no affected value is to be placed in issued-for-approval documents without a published-source check.
3. Do not apply the 1705.3 **three stories or less** concrete-footing waivers, or the 1705.13 **10.5 m / 7.5 m** seismic-SI waivers, to this high-rise.
4. Do not follow commentary that merges 1705.18 high-rise with Risk Category III/IV, or that restates interior-wall weight as **720 N/m²**.
5. Do not import SBC 304/305/306, AISC 341, ASTM E2174/E2393/E605, Section 909, Section 403.2.3 or Chapter 16 V/SDC values.
6. Record AHJ, SDC, wind, construction-type and fire-strategy decisions in the project Golden Thread; this matrix is not evidence of SCD NOC, structural observation close-out, or stamped compliance.

16. Coverage summary

Internal inventory of the attached Chapter 17 extract (numbered code, exceptions, tables, footnotes; commentary excluded). Row-level records are not published.

- **Inventory scope:** numbered code, exceptions, tables, footnotes (commentary excluded)
- **Total independently checkable numeric records:** 156

- **Verified:** 106
- **Verify source:** 50
- **Numeric records in Sections 1701, 1702, 1703, 1706 and 1707:** 0

Counts by top-level section

Top-level section	Records
1701	0
1702	0
1703	0
1704	4
1705 (body, excluding appended tables)	87
1706	0
1707	0
1708	3
1709	13

Appended-table coverage

Appended table	Records	Verify source records
Table 1705.2.3	3	3
Table 1705.3	22	22
Table 1705.5.3	9	9
Table 1705.6	5	5
Table 1705.7	4	4
Table 1705.8	2	2
Table 1705.13.7	4	4

Coverage cross-check against SBC 201 Chapter 17 Special Inspections and Tests (2024)_CS.md was topics-only; owner-employed SI and SSI; high-rise structural observation; concrete/steel SI; firestop SI in high-rise / RC III–IV / Group R OL > 250. No CS.md value was copied into a matrix cell.

17. Unresolved-source register

Hold points for the 50 **Verify source** inventory records. Counts are record counts, not distinct numeric values. No value in this register is a design-release figure.

Affected table / clause	Why unverified	Control note
1705.3 Exception items 2–5	Flattened list; items 3–5 appear un-nested; <i>f</i> <i>c</i> printed as 17. MPa	High-rise fails the three-storey gate; do not reconstruct IBC 2.1/2.2/2.3 numbering from memory; verify published exception if a separate low-rise structure is claimed
Table 1705.2.3	3 concatenated continuous/periodic cells	Verify published joist table before assigning periodic SI
Table 1705.3	20 concatenated C/P marks, Item 11 SDC text, and Item 2b “maximum 7.8 mm ” fillet-weld token	Do not repair 7.8 mm to another millimetre; verify published table before SSI frequencies
Table 1705.5.3 (cite also printed Table 1/705.5.3)	9 concatenated C/P cells	Mass timber only; verify published table if Types IV-A/B/C are used
Table 1705.6	5 concatenated C/P cells	Use 1705.6 charging text and the 90 percent density exception; verify table frequencies
Table 1705.7	4 concatenated continuous marks (items 1–4)	Verify published driven-pile table if that system is selected
Table 1705.8	2 concatenated continuous marks	Verify published CIP-pile table if that system is selected
Table 1705.13.7	4 concatenated periodic marks	Verify published rack table only if 2400 mm racks in SDC D–F exist
Unnumbered wood SFRS SI (after 1705.13.1.2)	Subsection heading missing at page 1596 split	Mandatory SDC C–F / 100 mm values retained; verify printed 1705.13.2 number
1705.14.3	Subsection absent (1705.14.2 jumps to 1705.14.4) while 1704.5 Item 3 cites it	No designated-system test values adopted
1705.15 charging heading	Section number missing before “Sprayed fire-resistant materials”	1705.15.1–1705.15.6 values retained; verify printed 1705.15 number
Unnumbered AAMA window labeling (after 1709.5)	1709.5.1 heading missing	Labeling rule retained; verify printed subsection number